

# SIMATS ENGINEERING

## ASSESSMENT TOOL 2

**COURSE NAME: INTERNET PROGRAMMING**  
**COURSE CODE: CSA4305**

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### COURSE OUTCOME COVERED

**CO -2** : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior.  
(L3)

Assessment Tool : Design Task

Weightage: 20%

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### QUESTIONS:

**Total Marks: 20**

#### Question 1: Responsive Navigation Bar Design - Marks(4)

Suppose that an e-commerce website requires a responsive navigation bar that adapts across desktop, tablet, and mobile devices.

- (a) Identify key components required in a navigation bar for an e-commerce website.
  - (b) Design a responsive navigation bar using appropriate CSS techniques.
  - (c) Demonstrate how the navigation adapts to smaller screens (e.g., hamburger menu).
  - (d) Evaluate the usability and suggest improvements for better user experience.
- 

#### Question 2: Product Hover Effect - Marks(4)

Suppose that an e-commerce website wants to enhance user interaction through visual effects on product items.

- (a) Identify suitable CSS properties for hover effects.
  - (b) Suggest appropriate **CSS animation or transition** for product interaction.
  - (c) Design a hover effect for a product card.
  - (d) Evaluate performance and user experience considerations.
- 

#### Question 3: Layout Selection (Flexbox vs Grid) - Marks(4)

Suppose that a dashboard layout needs to be designed for a web application.

- (a) Identify layout requirements of a dashboard (rows, columns, alignment).
  - (b) Compare **Flexbox and Grid** based on layout needs.
  - (c) Choose the most suitable layout technique and justify your choice.
  - (d) Evaluate the scalability and responsiveness of the chosen approach.
- 

#### Question 4: Mobile-First Blog Layout - Marks(4)

Suppose that a blog page must be designed following a mobile-first approach.

- (a) Identify key sections of a blog layout.
  - (b) Design a mobile-first layout using CSS.
  - (c) Demonstrate how the layout scales for larger screens using media queries.
  - (d) Evaluate accessibility and readability of the design.
- 

#### Question 5: JavaScript Event Handling for Dynamic Menu - Marks(4)

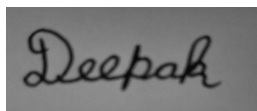
Suppose that a website includes a dynamic menu that responds to user interactions.

- (a) Identify appropriate JavaScript events for menu interaction.
- (b) Design event handling logic for menu toggle functionality.
- (c) Implement interaction for user actions (click/hover).
- (d) Evaluate performance and suggest improvements.

#### **Code of Conduct:**

*I Deepak V-192411021 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

#### **Signature of the student:**

A rectangular box containing a handwritten signature in black ink. The signature appears to be 'Deepak' written in a cursive, slightly slanted script.

## RUBRICS FOR CALCULATION:

| Criteria                         | Weightage (%) | Level 4 – Exemplary (4)                                                 | Level 3 – Proficient (3)                 | Level 2 – Developing (2)                  | Level 1 – Beginning (1)                  |
|----------------------------------|---------------|-------------------------------------------------------------------------|------------------------------------------|-------------------------------------------|------------------------------------------|
| <b>Understanding of Concept</b>  | 25            | Demonstrates complete understanding of design requirements and concepts | Good understanding with minor gaps       | Basic understanding with some errors      | Poor understanding or incorrect concepts |
| <b>Design / Implementation</b>   | 30            | Well-structured, efficient, and responsive design implementation        | Correct implementation with minor issues | Basic implementation with inconsistencies | Incorrect or incomplete implementation   |
| <b>Use of Tools (CSS/JS)</b>     | 20            | Effective and advanced use of tools/techniques                          | Appropriate use of tools                 | Limited or inconsistent use               | Incorrect or minimal use                 |
| <b>Analysis &amp; Evaluation</b> | 15            | Insightful analysis with strong justification and optimization          | Good analysis with justification         | Basic analysis with limited depth         | Poor or no analysis                      |
| <b>Documentation / Clarity</b>   | 10            | Clear, well-structured, and properly presented solution                 | Mostly clear with minor issues           | Basic clarity, missing details            | Poor or unclear documentation            |

## Question 1: Responsive Navigation Bar Design

CODE:

```
<!DOCTYPE html>
<html lang="en">

<head>

<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Responsive Navigation Bar</title>

<style>

*{
margin:0;
padding:0;
box-sizing:border-box;
font-family:Arial,sans-serif;
}

body{
background:#f4f4f4;
}

nav{
display:flex;
justify-content:space-between;
align-items:center;
background:#333;
padding:15px 25px;
color:white;
}

.logo{
font-size:24px;
font-weight:bold;
}

.menu{
display:flex;
list-style:none;
gap:20px;
}

.menu li a{
text-decoration:none;
```

```
color:white;
font-size:18px;
transition:0.3s;
}
```

```
.menu li a:hover{
color:#00d4ff;
}
```

```
.hamburger{
display:none;
font-size:30px;
cursor:pointer;
}
```

```
@media(max-width:768px){
```

```
.menu{
display:none;
flex-direction:column;
width:100%;
background:#444;
position:absolute;
top:65px;
left:0;
padding:20px;
}
```

```
.menu.active{
display:flex;
}
```

```
.hamburger{
display:block;
}
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<nav>
```

```
<div class="logo">
ShopEasy
```

```
</div>
```

```
<ul class="menu" id="menu">
```

```
<li><a href="#">Home</a></li>
```

```
<li><a href="#">Products</a></li>
```

```
<li><a href="#">Categories</a></li>
```

```
<li><a href="#">Cart</a></li>
```

```
<li><a href="#">Contact</a></li>
```

```
</ul>
```

```
<div class="hamburger" onclick="toggleMenu()">
```



```
</div>
```

```
</nav>
```

```
<div style="padding:40px;text-align:center;">
```

```
<h1>Responsive Navigation Bar</h1>
```

```
<p>
```

Resize the browser window to observe how the navigation bar adapts to desktop, tablet, and mobile devices.

```
</p>
```

```
</div>
```

```
<script>
```

```
function toggleMenu(){
```

```
document.getElementById("menu").classList.toggle("active");
```

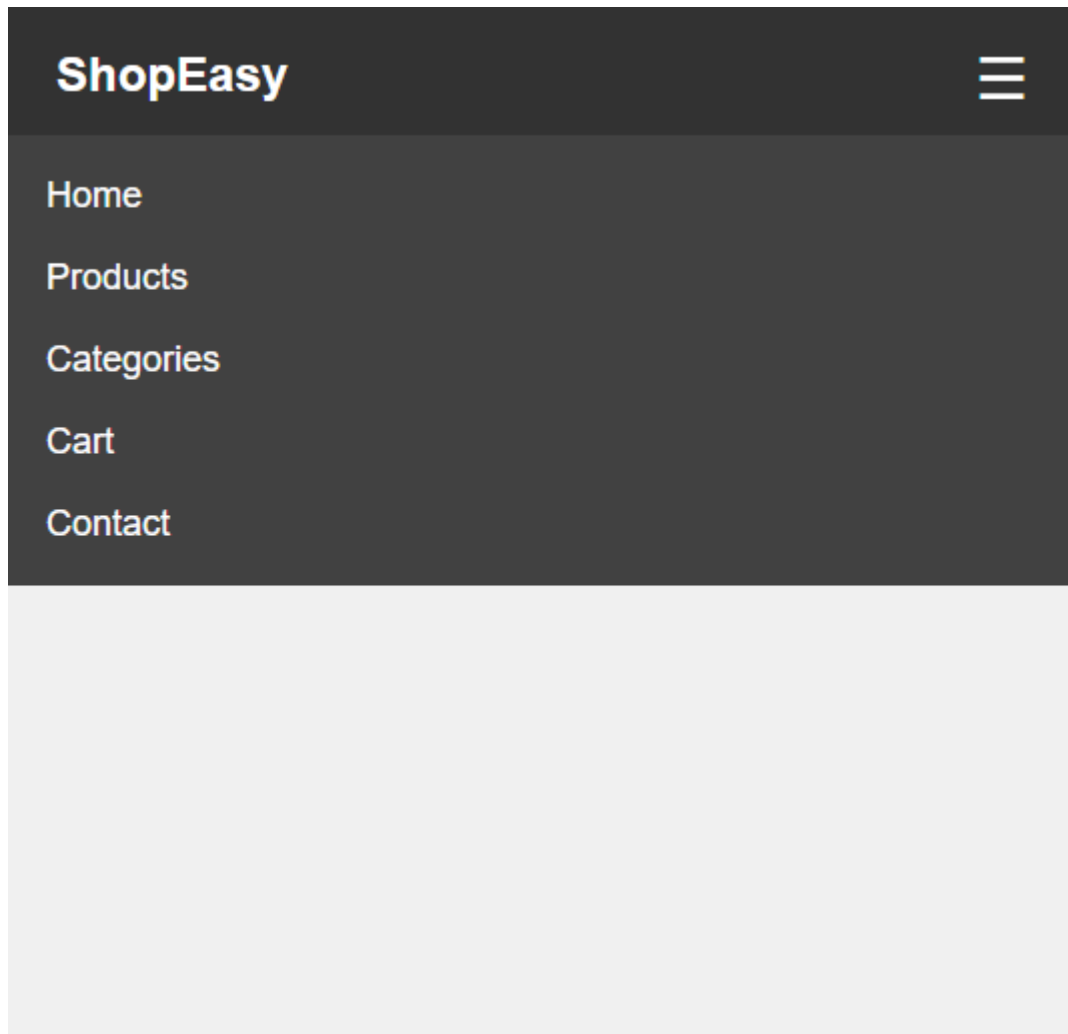
```
}
```

```
</script>
```

```
</body>
```

```
</html>
```

OUTPUT:



## Question 2: Product Hover Effect

CODE:

```
<!DOCTYPE html>  
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>Product Hover Effect</title>
```

```
<style>
```

```
{  
margin:0;  
padding:0;
```

```
box-sizing:border-box;
font-family:Arial,sans-serif;
}
```

```
body{
background:#f2f2f2;
display:flex;
justify-content:center;
align-items:center;
height:100vh;
}
```

```
.product-card{
width:300px;
background:white;
border-radius:10px;
overflow:hidden;
box-shadow:0 4px 10px rgba(0,0,0,0.2);
text-align:center;
transition:transform 0.4s ease, box-shadow 0.4s ease;
cursor:pointer;
}
```

```
.product-card:hover{
transform:translateY(-10px) scale(1.05);
box-shadow:0 10px 20px rgba(0,0,0,0.3);
}
```

```
.product-card img{
width:100%;
height:220px;
object-fit:cover;
transition:transform 0.4s ease;
}
```

```
.product-card:hover img{
transform:scale(1.1);
}
```

```
.product-info{
padding:20px;
}
```

```
.product-info h2{
margin-bottom:10px;
color:#333;
}
```



```
.product-info p{
font-size:18px;
color:green;
margin-bottom:15px;
}
```

```
button{
padding:10px 20px;
border:none;
background:#007BFF;
color:white;
font-size:16px;
border-radius:5px;
cursor:pointer;
transition:background 0.3s;
}
```

```
button:hover{
background:#0056b3;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class="product-card">
```

```

```

```
<div class="product-info">
```

```
<h2>Wireless Headphones</h2>
```

```
<p>₹2,999</p>
```

```
<button>Add to Cart</button>
```

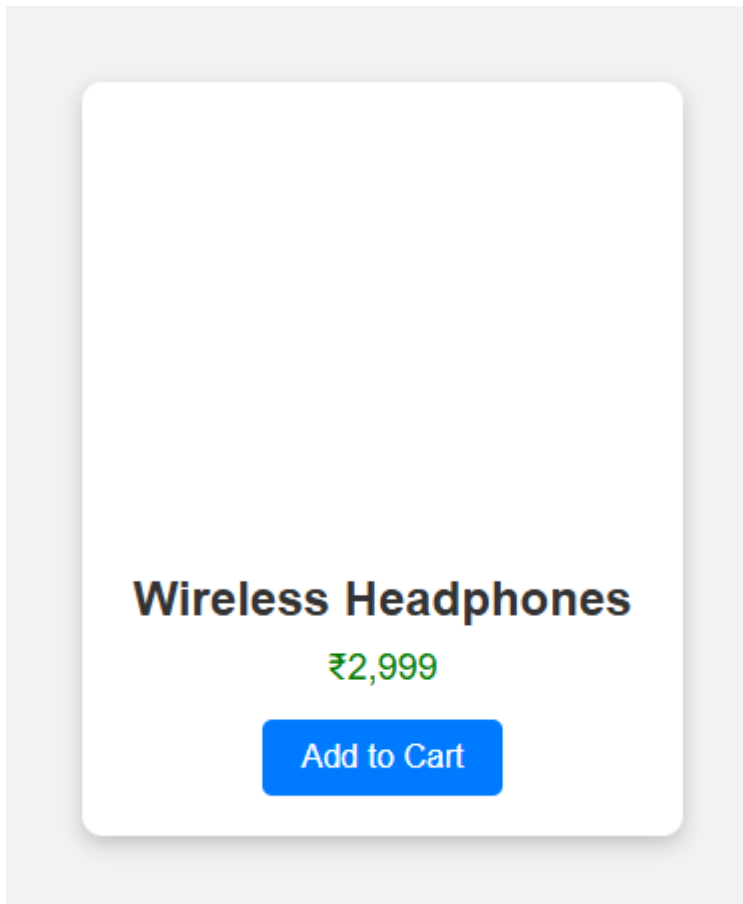
```
</div>
```

```
</div>
```

```
</body>
```

```
</html>
```

OUTPUT:



### Question 3: Layout Selection (Flexbox vs Grid)

CODE:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>Dashboard Layout - CSS Grid</title>
```

```
<style>
```

```
*{  
margin:0;  
padding:0;  
box-sizing:border-box;  
font-family:Arial,sans-serif;  
}
```

```
body{
background:#f4f4f4;
}
```

```
.dashboard{
display:grid;
grid-template-columns:250px 1fr;
grid-template-rows:70px 1fr 60px;
grid-template-areas:
"header header"
"sidebar main"
"footer footer";
min-height:100vh;
}
```

```
.header{
grid-area:header;
background:#2c3e50;
color:white;
display:flex;
justify-content:center;
align-items:center;
font-size:28px;
}
```

```
.sidebar{
grid-area:sidebar;
background:#34495e;
color:white;
padding:20px;
}
```

```
.sidebar ul{
list-style:none;
}
```

```
.sidebar li{
padding:15px 0;
border-bottom:1px solid rgba(255,255,255,0.2);
}
```

```
.main{
grid-area:main;
padding:20px;
display:grid;
grid-template-columns:repeat(auto-fit,minmax(220px,1fr));
gap:20px;
}
```

```
.card{
background:white;
padding:30px;
border-radius:8px;
box-shadow:0 4px 8px rgba(0,0,0,0.2);
text-align:center;
font-size:20px;
}
```

```
.footer{
grid-area:footer;
background:#2c3e50;
color:white;
display:flex;
justify-content:center;
align-items:center;
}
```

```
@media(max-width:768px){
```

```
.dashboard{
```

```
grid-template-columns:1fr;
```

```
grid-template-areas:
```

```
"header"
```

```
"sidebar"
```

```
"main"
```

```
"footer";
```

```
}
```

```
.main{
```

```
grid-template-columns:1fr;
```

```
}
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class="dashboard">
```

```
<div class="header">
```

```
Dashboard
```

```
</div>
```

```
<div class="sidebar">
```

```
<h2>Menu</h2>
```

```
<ul>
```

```
<li>Home</li>
```

```
<li>Analytics</li>
```

```
<li>Users</li>
```

```
<li>Settings</li>
```

```
<li>Reports</li>
```

```
</ul>
```

```
</div>
```

```
<div class="main">
```

```
<div class="card">Sales</div>
```

```
<div class="card">Revenue</div>
```

```
<div class="card">Orders</div>
```

```
<div class="card">Customers</div>
```

```
<div class="card">Products</div>
```

```
<div class="card">Feedback</div>
```

```
</div>
```

```
<div class="footer">
```

## Dashboard Layout Using CSS Grid

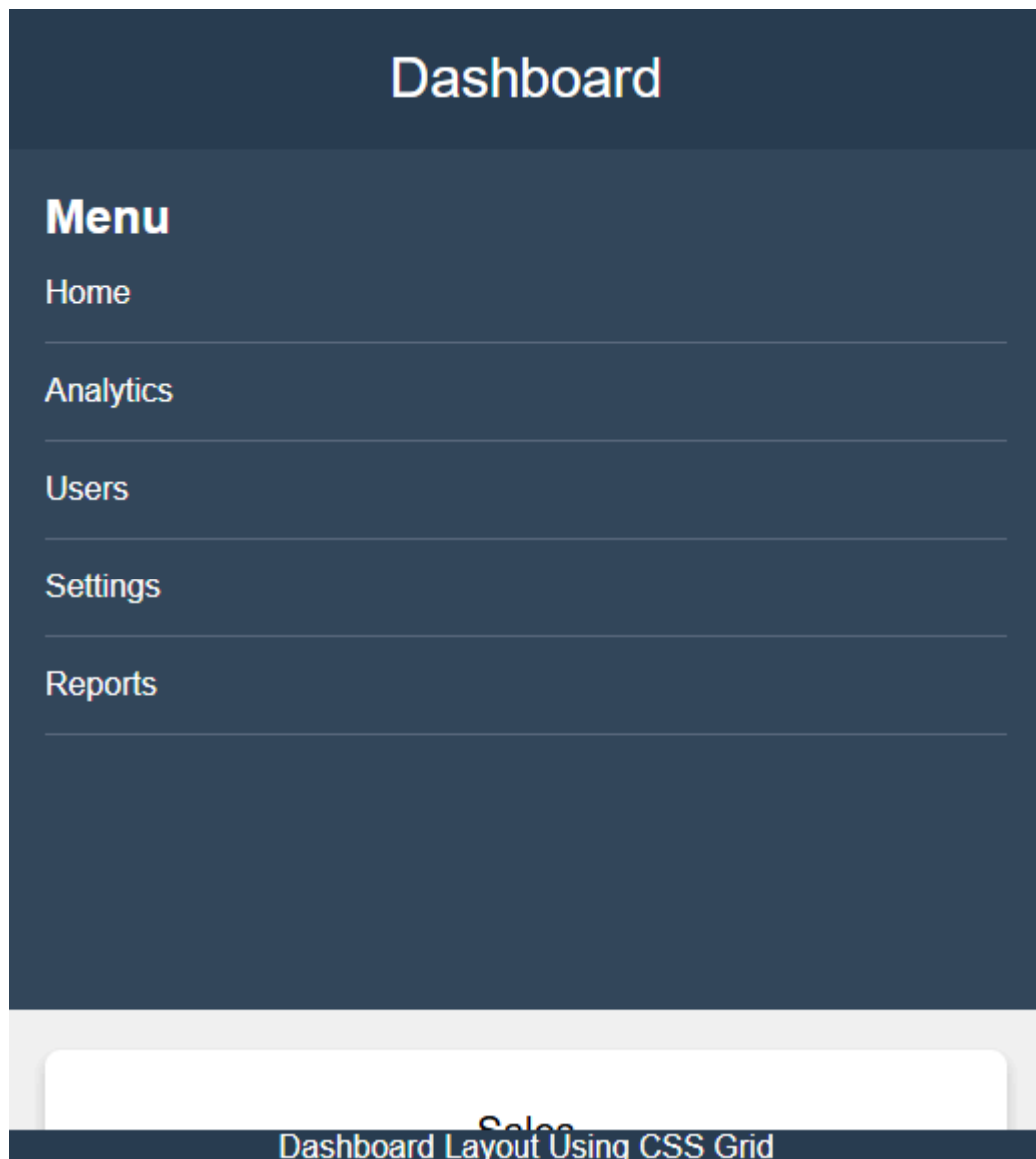
```
</div>
```

```
</div>
```

```
</body>
```

```
</html>
```

OUTPUT:



## Question 4: Mobile-First Blog Layout

CODE:

```
<!DOCTYPE html>
<html lang="en">

<head>

<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Mobile-First Blog Layout</title>

<style>

*{
margin:0;
padding:0;
box-sizing:border-box;
font-family:Arial,sans-serif;
}

body{
background:#f4f4f4;
line-height:1.6;
}

header{
background:#2c3e50;
color:white;
padding:20px;
text-align:center;
}

nav{
background:#34495e;
display:flex;
flex-direction:column;
text-align:center;
}

nav a{
color:white;
text-decoration:none;
padding:15px;
border-bottom:1px solid rgba(255,255,255,0.2);
}
```

```
.container{
display:flex;
flex-direction:column;
padding:20px;
gap:20px;
}
```

```
article{
background:white;
padding:20px;
border-radius:8px;
box-shadow:0 4px 8px rgba(0,0,0,0.2);
}
```

```
article h2{
margin-bottom:10px;
}
```

```
aside{
background:#ecf0f1;
padding:20px;
border-radius:8px;
}
```

```
footer{
background:#2c3e50;
color:white;
text-align:center;
padding:20px;
margin-top:20px;
}
```

```
@media(min-width:768px){
```

```
nav{
flex-direction:row;
justify-content:center;
}
```

```
nav a{
border:none;
padding:20px;
}
```

```
.container{
flex-direction:row;
}
```



```
article{
flex:3;
}
```

```
aside{
flex:1;
}
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<header>
```

```
<h1>Tech Blog</h1>
```

```
<p>Mobile-First Responsive Blog Layout</p>
```

```
</header>
```

```
<nav>
```

```
<a href="#">Home</a>
```

```
<a href="#">Articles</a>
```

```
<a href="#">Categories</a>
```

```
<a href="#">About</a>
```

```
<a href="#">Contact</a>
```

```
</nav>
```

```
<div class="container">
```

```
<article>
```

```
<h2>Latest Technology Trends</h2>
```

```
<p>
```

This blog demonstrates a mobile-first responsive layout. On mobile devices, the content is displayed in a single column for better readability. On tablets and desktops, the layout automatically expands into multiple columns using

CSS media queries.

</p>

</article>

<aside>

<h3>Recent Posts</h3>

<ul>

<li>Artificial Intelligence</li>

<li>Cloud Computing</li>

<li>Cyber Security</li>

<li>Web Development</li>

<li>Machine Learning</li>

</ul>

</aside>

</div>

<footer>

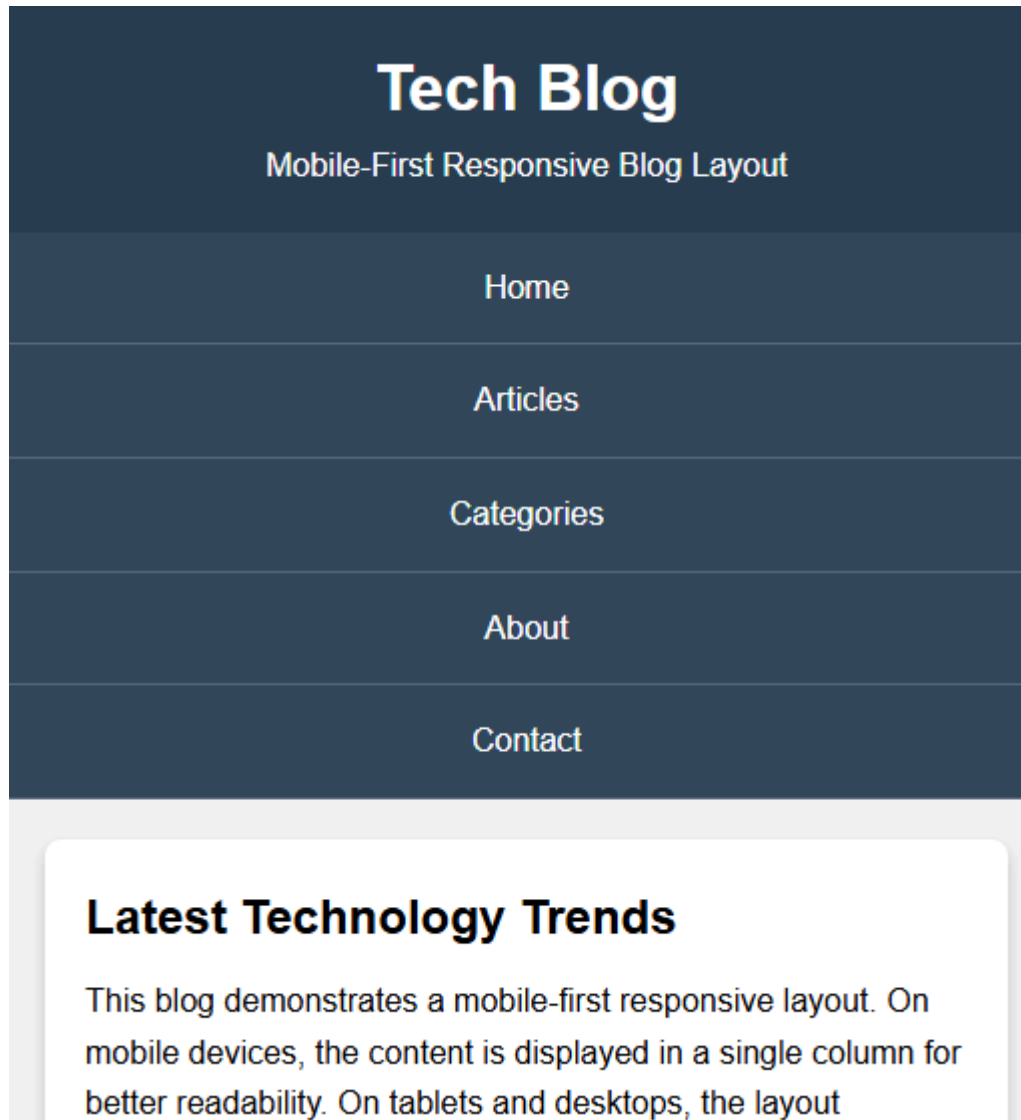
<p>&copy; 2026 Tech Blog. All Rights Reserved.</p>

</footer>

</body>

</html>

OUTPUT:



### Question 5: JavaScript Event Handling for Dynamic Menu

CODE:

```
<!DOCTYPE html>
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>Dynamic Menu Using JavaScript</title>
```

```
<style>
```

```
*{
```

```
margin:0;
padding:0;
box-sizing:border-box;
font-family:Arial,sans-serif;
}
```

```
body{
background:#f4f4f4;
}
```

```
header{
background:#2c3e50;
color:white;
padding:20px;
display:flex;
justify-content:space-between;
align-items:center;
}
```

```
.logo{
font-size:24px;
font-weight:bold;
}
```

```
.menu-btn{
font-size:30px;
cursor:pointer;
display:none;
}
```

```
nav{
background:#34495e;
}
```

```
nav ul{
display:flex;
justify-content:center;
list-style:none;
}
```

```
nav ul li{
padding:18px;
}
```

```
nav ul li a{
color:white;
text-decoration:none;
font-size:18px;
```

```
transition:0.3s;
}
```

```
nav ul li:hover{
background:#1abc9c;
}
```

```
@media(max-width:768px){
```

```
.menu-btn{
display:block;
}
```

```
nav{
display:none;
}
```

```
nav.active{
display:block;
}
```

```
nav ul{
flex-direction:column;
text-align:center;
}
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<header>
```

```
<div class="logo">
My Website
</div>
```

```
<div class="menu-btn" onclick="toggleMenu()">
≡
</div>
```

```
</header>
```

```
<nav id="menu">
```

```
<ul>

<li><a href="#">Home</a></li>

<li><a href="#">Services</a></li>

<li><a href="#">Products</a></li>

<li><a href="#">Gallery</a></li>

<li><a href="#">Contact</a></li>

</ul>

</nav>

<div style="padding:40px;text-align:center;">

<h1>Dynamic Navigation Menu</h1>

<p>
Click the hamburger icon on smaller screens to open and close the navigation menu.
Hover over the menu items to see the interactive effect.
</p>

</div>

<script>

function toggleMenu(){

document.getElementById("menu").classList.toggle("active");

}

</script>

</body>

</html>
```

OUTPUT:

